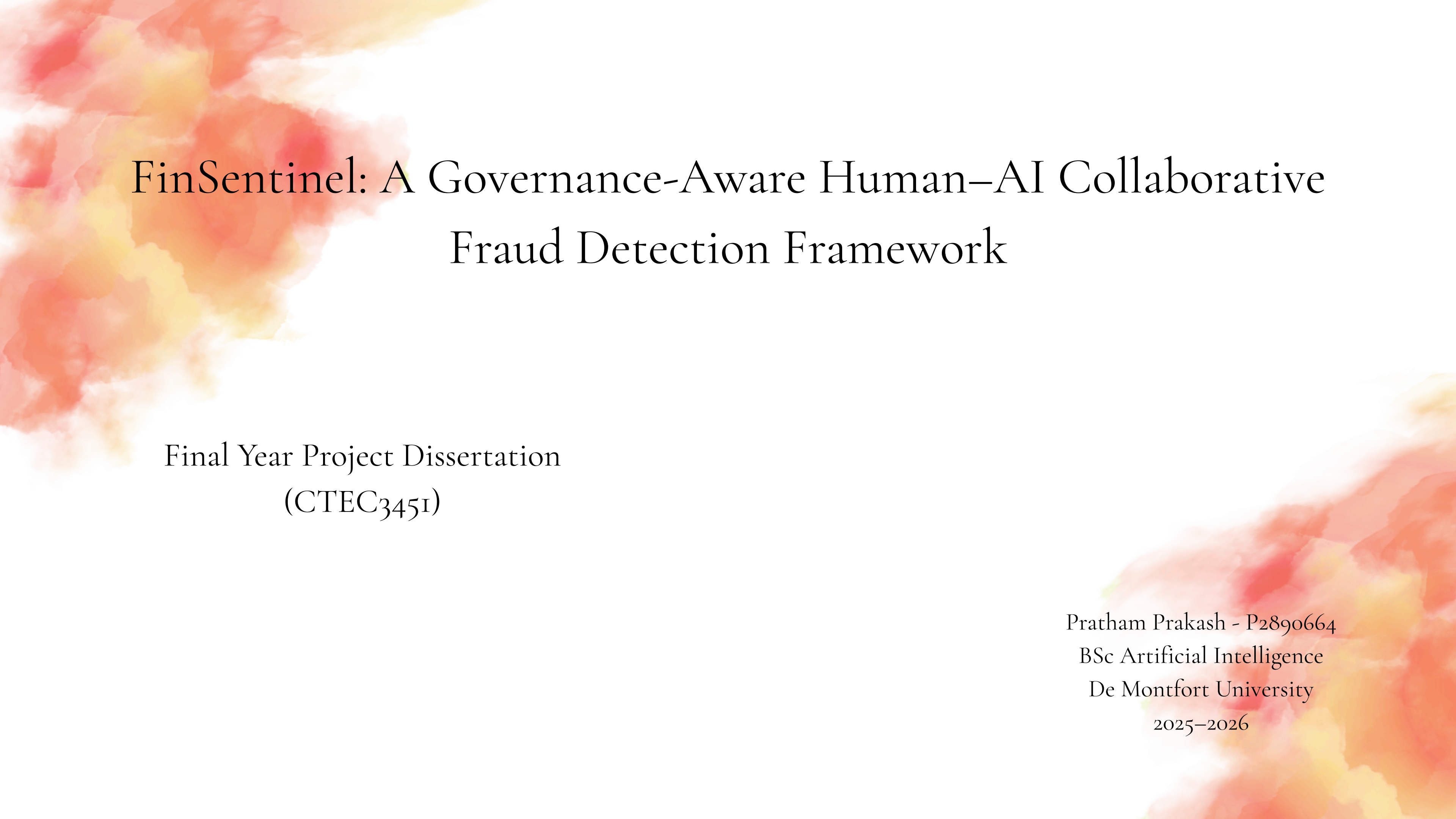




# FinSentinel: A Governance-Aware Human–AI Collaborative Fraud Detection Framework

Final Year Project Dissertation  
(CTEC345I)

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# Problem Statement & Research Gap

## Current Challenges in Fraud Detection

- Black-box decision behaviour
- Limited operational governance
- Limited human oversight
- Evolving fraud behaviour
- Limited long-term reliability

## Research Gap

Existing approaches often investigate:

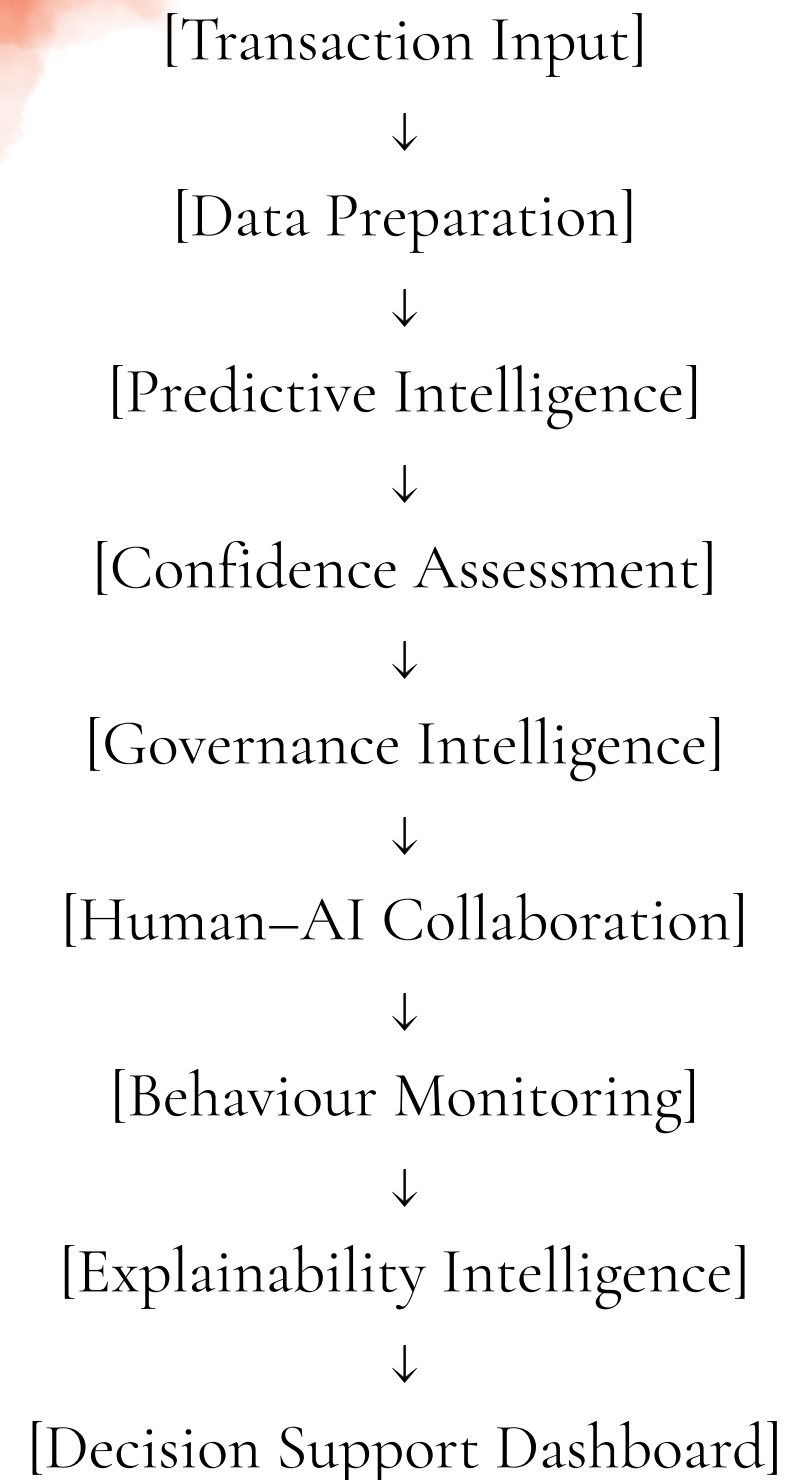
- Predictive models
- Explainability methods
- Human–AI collaboration
- Drift detection

Independently rather than within a unified framework

## Project Aim:

Develop a governance-aware explainable Human–AI collaborative fraud detection framework integrating: Predictive Intelligence, Operational Routing, Human Oversight, Behavioural Drift Intelligence

# FinSentinel Framework Architecture



## Integrated Components

- Predictive Intelligence
- Governance Intelligence
- Explainability Intelligence
- Behavioural Monitoring
- Human Oversight



# Dataset & Data Preparation

## Dataset

- Credit Card Fraud Detection Dataset - Kaggle
- Total Transactions: 284,807
- Fraud Cases: 492
- Highly imbalanced dataset

## Preprocessing Pipeline

- Missing value inspection
- Numeric feature filtering
- Leakage detection
- Temporal ordering using Time variable
- Train / Validation / Test split
- StandardScaler (LR & MLP only)
- Weighted Scaling (Scale\_pro\_weights)

## Standardisation

$$z = (x - \mu) / \sigma$$

where:

- $x$  = feature value
- $\mu$  = mean
- $\sigma$  = standard deviation



# Predictive Intelligence Implementation

## Implemented Models

Classical Model

- Logistic Regression

Ensemble Models

- Random Forest
- XGBoost
- LightGBM
- CatBoost

Neural Model

- Multi-Layer Perceptron (MLP)

## Model Evaluation Metrics

- Precision
- Recall
- F1-score
- PR-AUC
- ROC-AUC
- Brier Score\*

## Prediction Output

Input Features



Machine Learning Model



Fraud Probability

**Selected Operational Model - Random Forest**

### **Reason:**

Strong predictive performance, Robust behaviour under imbalanced data, Suitable for operational deployment

# Confidence Assessment

## Probability Calibration

### Calibration:

- Method = Sigmoid (Platt Scaling)
- Cross Validation = 3-fold

### Purpose:

- Improve probability reliability
- Align predicted confidence with observed outcomes
- Support downstream governance decisions

## Threshold Optimisation

### Implemented using:

- Threshold range: 0.01–0.99
- Total thresholds evaluated: 1000
- Selection criterion: Maximum F1-score

$$F1 = (2 \times (Precision \times Recall) \div (Precision + Recall))$$

## Threshold Selection Formula Model Training

### For each threshold $t$ :

Probability  $\rightarrow$  Prediction



Calculate F1 score



Store  $F1(t)$



Best Threshold

$$t^* = \operatorname{argmax}(F1(t))$$

Threshold search:

0.01–0.99

(1000 candidate thresholds)

# Governance Intelligence & Human–AI Collaboration

## Governance Strategy Configuration

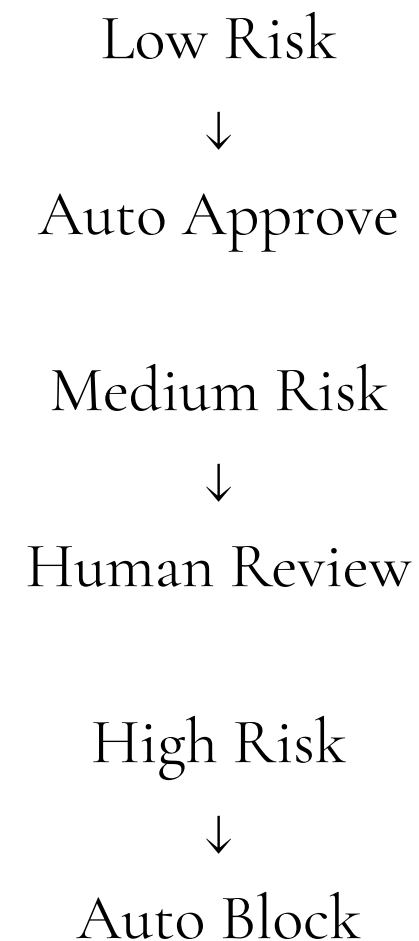
### Purpose:

- Convert prediction confidence into operational decisions
- Balance fraud detection and analyst workload
- Support different governance strategies

### Strategies:

- Balanced Governance
- Conservative Governance
- Aggressive Fraud Containment
- Operational Efficiency

## Human–AI Routing Logic



## Routing Logic

```
if p < approve_threshold:  
    Auto Approve  
  
elif p >= block_threshold:  
    Auto Block  
  
else:  
    Human Review
```



# Behavioural Drift Intelligence

## Drift Detection Method

### Algorithm:

Page–Hinkley

### Purpose:

- Monitor changing transaction behaviour
- Detect behavioural instability
- Support long-term operational reliability

### Parameters Used:

- `min_instances` = 50 - ignore
- `delta` = 0.02 - max detectable change
- `threshold` = 50 - accumulated evidence

## Adaptive Governance Response

Low Drift

No threshold adjustment

Moderate Drift

Threshold Adjustment = -0.05

High Drift

Threshold Adjustment = -0.10

## Behavioural Monitoring Workflow

Transaction Behaviour

↓

Page–Hinkley Detection

↓

Drift Score

↓

Governance Adjustment

# Explainability Intelligence

## SHAP Integration

### Purpose:

- Improve prediction transparency
- Explain model behaviour
- Support analyst interpretation
- Increase operational trust

### Explainers Used

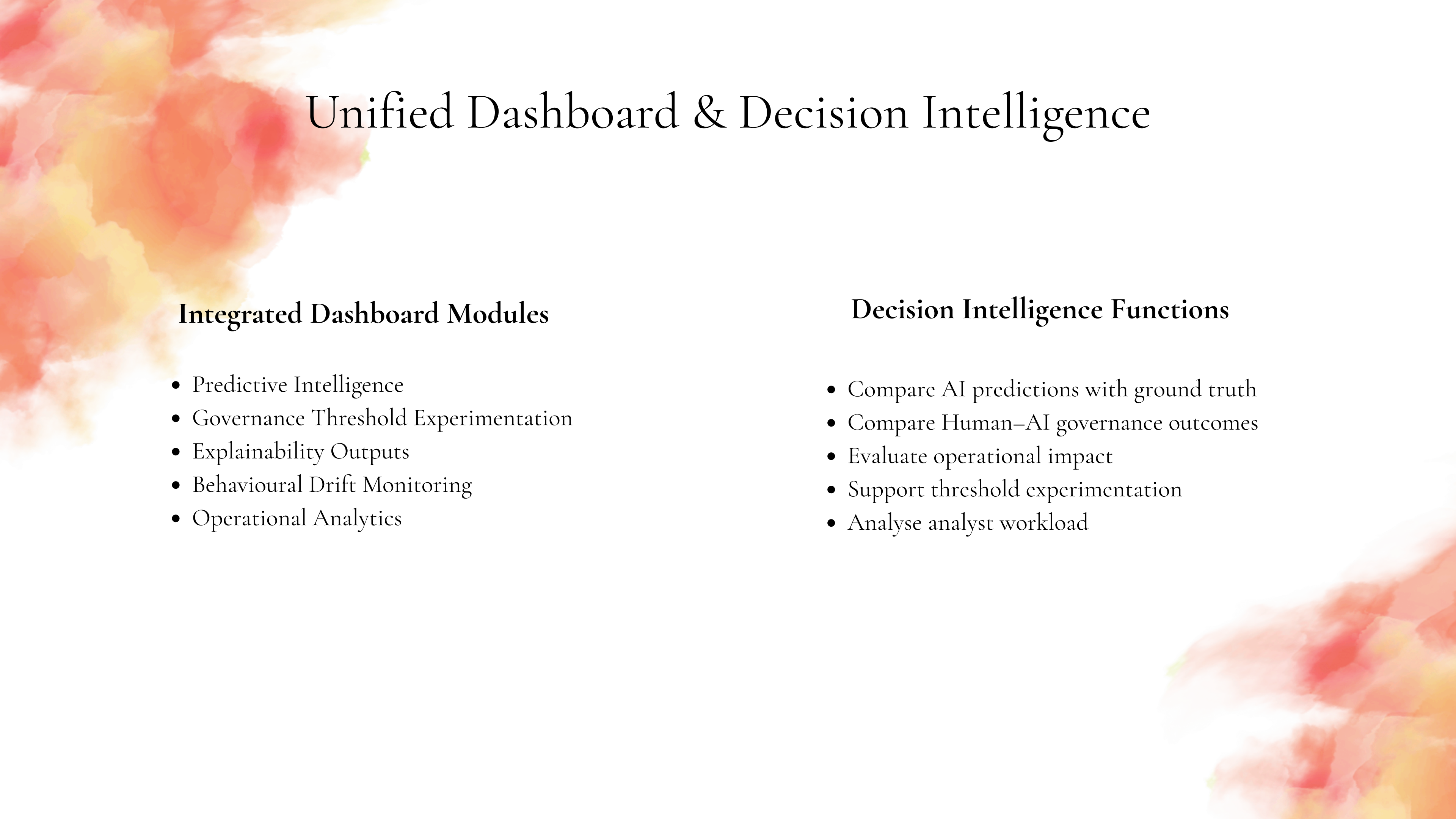
- Tree Models - TreeExplainer
- Linear Models - LinearExplainer
- Neural Models - KernelExplainer

## Explainability Outputs

- Global Feature Importance
- Local Prediction Explanations
- SHAP Waterfall Analysis
- Feature Contribution Analysis
- Analyst Narrative Support

## SHAP Prediction Logic

Final Prediction = Base Value +  $\sum$  Feature Contributions



# Unified Dashboard & Decision Intelligence

## Integrated Dashboard Modules

- Predictive Intelligence
- Governance Threshold Experimentation
- Explainability Outputs
- Behavioural Drift Monitoring
- Operational Analytics

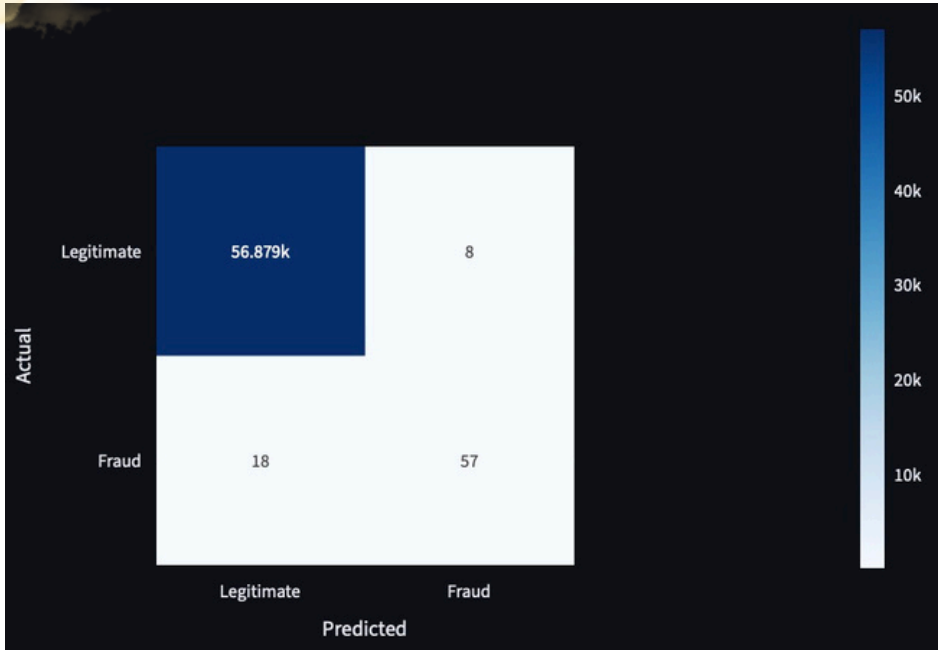
## Decision Intelligence Functions

- Compare AI predictions with ground truth
- Compare Human–AI governance outcomes
- Evaluate operational impact
- Support threshold experimentation
- Analyse analyst workload



# Results & Evaluation

AI Confusion Matrix



Governance Routing Matrix



## Key Findings

- Selective escalation improved fraud handling while introducing operational cost trade-offs
- Human review reduced missed fraud risk
- Wrongful auto-blocking remained minimal
- Trade-off observed between fraud capture and operational cost
- Increased intervention → increased analyst workload

| Metric             | AI Only | Human-AI |
|--------------------|---------|----------|
| Fraud detected     | 57      | 72       |
| Fraud missed       | 18      | 3        |
| Human intervention | 0       | 8420     |

# Research Question Evaluation

|      | Research Question                                                             | Evaluation Outcome                            |
|------|-------------------------------------------------------------------------------|-----------------------------------------------|
| RQ1: | Can governance-aware AI improve operational trust in fraud detection systems? | Governance improved operational trust         |
| RQ2: | Can selective escalation reduce operational risk?                             | Selective escalation reduced operational risk |
| RQ3: | Can explainability improve decision transparency?                             | SHAP improved decision transparency           |
| RQ4: | Can behavioural drift intelligence improve long-term operational reliability? | Drift monitoring improved reliability         |

**Overall Finding:**  
FinSentinel successfully addressed all research objectives.

# Contributions & Future Work

## Project Contributions

- Human–AI collaborative governance framework
- Integration of predictive intelligence, explainability and behavioural monitoring
- Threshold-based governance routing for operational decision support
- Probability calibration for confidence reliability
- Unified dashboard for operational visualisation

## Key Contribution

FinSentinel extends beyond prediction-focused systems by combining predictive intelligence with governance, explainability, behavioural monitoring and human oversight.

## Future Work

- Real-time transaction streaming
- Dynamic threshold adaptation
- Analyst feedback learning
- Reinforcement-based decision adaptation
- Large Language Model integration
- More extensive hyperparameter optimisation across predictive models
- Learnable governance strategy optimisation rather than fixed configurations
- Adaptive drift response using data-driven threshold adjustment
- Integrated calibration-aware decision optimisation
- Reinforcement learning for dynamic governance adaptation





THANK YOU!



# MODEL RUN & QUESTIONS\*